

Year 7 · Chapter 2 · Angles: answers (teacher copy)

2.01 Naming angles (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: $\angle ABC$ or $\angle CBA$
- Ex 1b: vertex K in the middle
- Ex 2a: $\angle QOR$
- Ex 2b: $\angle QOR$, $\angle ROP$, $\angle QOP$
- Ex 3a: $\angle LON$
- Ex 3b: There are several angles at B, so we need 3 letters

Easy · Set A

- $\angle PQR$
- $\angle DEF$
- $\angle XYZ$
- $\angle MNK$
- $\angle STU$
- $\angle GHJ$

Easy · Set A2 (quick drill)

- B
- Q
- Y
- E
- N
- T
- H
- K

Easy · Set B

- vertex P, arms PF and PR
- vertex Z, arms ZM and ZQ

Medium · Set C

- $\angle DBE$
- $\angle DBE$
- $\angle CBA$
- 6
- $\angle DBA$
- $\angle CBE$

Challenging · Set D

- $\angle LMN$
- $\angle QRS$
- Yes: the vertex B is in the middle of both
- e.g. two angles sharing arm QR

Extension

- 10
- $\angle AOB$, $\angle AOC$, $\angle AOD$, $\angle BOC$, $\angle BOD$, $\angle COD$

Exit ticket

- Level 1: $\angle TUV$
- Level 2: U, because it is the point where the arms meet
- Level 3: two angles sharing arm BC

2.02 Measuring and drawing angles

(booklet p. 9)

WE DO examples (teacher)

- Ex 1a: 64°
- Ex 1b: accurate 84° angle
- Ex 2a: 128°
- Ex 2b: accurate 117° angle
- Ex 3a: draw 150° on the other side
- Ex 3b: draw 60° on the other side

Easy · Set A

- 35°
- 72°
- 90°
- 105°
- 150°
- 48°

Easy · Set A2 (quick drill)

- 150°
- 60°
- 165°
- 90°
- 110°
- 30°
- 175°
- 120°

Easy · Set B

- 41° drawn
- 150° drawn

Medium · Set C

- 96° drawn
- 23° drawn
- 165° drawn
- 72° drawn

Challenging · Set D

- $360 - 195 = 165^\circ$ on the other side
- three-quarter turn
- 230°
- 215°

Extension

- 150° and 90°
- 75°

Exit ticket

- Level 1: 58°
- Level 2: accurate 135°
- Level 3: draw 110° the other way

2.03 Classifying angles (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: acute, right, obtuse, straight, reflex, revolution
- Ex 1b: $\angle WIH$, obtuse
- Ex 2a: obtuse
- Ex 2b: 270° , reflex
- Ex 3a: $\angle DRA$ acute (70°), $\angle ARB$ obtuse (110°)
- Ex 3b: No: 90° + more than 90° is already more than 180°

Easy · Set A (quick drill)

1. acute
2. right
3. obtuse
4. straight
5. reflex
6. acute
7. acute
8. obtuse
9. reflex
10. revolution
11. acute
12. obtuse
13. obtuse
14. reflex
15. reflex
16. acute

Easy · Set B

1. acute
2. obtuse
3. reflex
4. revolution
5. obtuse
6. reflex

Medium · Set C

1. $\angle GLU$ acute
2. $\angle VRP$ obtuse
3. $\angle PNE$ right
4. $\angle SMA$ reflex
5. $\angle ZXY$ straight
6. $\angle MVT$ acute

Challenging · Set D

1. 120° , obtuse
2. Yes, e.g. $60^\circ + 50^\circ = 110^\circ$
3. 100° , 125° , 150° , 175°
4. 19 (201° to 219°)

Extension

1. 900°
2. 3 revolutions

Exit ticket

1. Level 1: acute
2. Level 2: reflex; straight
3. Level 3: $\angle DRA$ acute; $\angle ARB$ obtuse

2.04 Complementary and supplementary angles (booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: 55° ; 12°
2. Ex 1b: 135° ; 80°
3. Ex 2a: $m = 62$
4. Ex 2b: $k = 133$
5. Ex 3a: $x = 29$
6. Ex 3b: 90° each

Easy · Set A (quick drill)

1. 55°
2. 12°
3. 86°
4. 30°
5. 45°
6. 1°
7. 65°
8. 22°
9. 78°
10. 9°
11. 150°
12. 65°
13. 8°
14. 116°
15. 90°
16. 45°

Easy · Set B

1. 135°
2. 80°
3. 2°
4. 90°
5. 168°
6. 45°

Medium · Set C

1. $m = 62$
2. $k = 133$
3. $a = 29$
4. $y = 58$
5. $p = 15$
6. $b = 90$

Challenging · Set D

1. 55° and 35°
2. 36°
3. $x = 40$
4. No: it is already more than 90°

Extension

1. 60°
2. 40°

Exit ticket

1. Level 1: 28°
2. Level 2: 62°
3. Level 3: $x = 45$

2.05 Angles at a point and vertically opposite angles (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: $a = 110$
2. Ex 1b: $m = 25$
3. Ex 2a: $a = 130$
4. Ex 2b: $x = 120$
5. Ex 3a: $p = 145$
6. Ex 3b: $p = 155$, $q = 25$, $r = 155$

Easy · Set A

1. $y = 145$
2. $a = 80$
3. $m = 44$
4. $x = 98$
5. $k = 25$
6. $d = 90$

Easy · Set A2 (quick drill)

1. 125°
2. 144°
3. 75°
4. 120°
5. 67°
6. 118°
7. 72°
8. 116°

Easy · Set B

1. $y = 35$
2. $m = 80$
3. $f = 60$

Medium · Set C

1. $f = 110$
2. $t = 83$
3. $x = 90$

Challenging · Set D

1. $x = 132$ (straight line), $y = 48$ (vertically opposite)
2. $t = 173$ (angles at a point)
3. 120° each
4. $115^\circ, 65^\circ, 115^\circ$

Extension

1. $30^\circ, 60^\circ, 90^\circ$
2. $x = 30$

Exit ticket

1. Level 1: 116
2. Level 2: 38°
3. Level 3: 135

2.06 Parallel and perpendicular lines

(booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: $AB \parallel DC$ (or $AD \parallel BC$); $AB \perp BC$
2. Ex 1b: AB is perpendicular to CD ; EF is parallel to GH
3. Ex 2a: construction
4. Ex 2b: construction
5. Ex 3a: construction
6. Ex 3b: e.g. shelves ($\parallel$), wall and floor ($\perp$)

Easy · Set A (quick drill)

1. DC
2. BC or AD
3. AD
4. No (not in a rectangle unless it is a square)
5. at B
6. $AD \parallel BC$

Easy · Set B

1. True
2. True
3. False
4. True

Medium · Set C

1. construction
2. construction

Challenging · Set D

1. construction
2. square construction

Extension

1. construction
2. every point on it is the same distance from A and B

Exit ticket

1. Level 1: $\parallel$
2. Level 2: e.g. wall and floor
3. Level 3: construction

2.07 Corresponding angles (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $a = 65$
3. Ex 2a: $m = 110$
4. Ex 2b: the angles sit in an F
5. Ex 3a: $x = 125, y = 55$
6. Ex 3b: Only when the lines are parallel

Easy · Set A

1. bottom ul
2. top lr
3. bottom ll
4. top ur
5. bottom lr
6. top ul

Easy · Set A2 (quick drill)

1. 65°
2. 110°
3. 38°
4. 142°
5. 90°
6. 75°
7. 120°
8. 51°

Easy · Set B

1. $a = 65$
2. $k = 50$
3. $d = 110$

Medium · Set C

1. $b = 35$
2. $m = 115$
3. $q = 118$ (corresponding 62° , then straight line)
4. $x = 122$
5. $y = 75$ (vertically opposite, then corresponding)
6. $t = 140$

Challenging · Set D

1. a line crossing two or more lines
2. drawing
3. No
4. angles on a straight line

Extension

1. $y = 40$
2. $a = 30$

Exit ticket

1. Level 1: 60
2. Level 2: an F
3. Level 3: $x = 20$

2.08 Alternate angles (booklet p. 45)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $x = 60$
3. Ex 2a: $k = 115$
4. Ex 2b: True
5. Ex 3a: $a = 48$, $b = 132$
6. Ex 3b: the angles sit inside a Z

Easy · Set A

1. bottom ul
2. top ll
3. bottom ur
4. top lr
5. bottom ul
6. top ll

Easy · Set A2 (quick drill)

1. 60°
2. 115°
3. 48°
4. 132°
5. 130°
6. 50°
7. 72°
8. 108°

Easy · Set B

1. $x = 70$
2. $p = 125$
3. $w = 38$

Medium · Set C

1. $c = 65$
2. $t = 112$
3. $d = 130$ (alternate 50° , straight line)
4. $e = 73$ (corresponding)
5. $g = 140$
6. $h = 100$ (alternate 80° , then straight line)

Challenging · Set D

1. drawing
2. $x = 45$
3. alternate angles and angles on a straight line
4. It is perpendicular to both lines ($y = 45$)

Extension

1. $a = 25$
2. $m = 35$; both 105°

Exit ticket

1. Level 1: 65
2. Level 2: Z
3. Level 3: $x = 35$

2.09 Co-interior angles (booklet p. 51)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $y = 60$
3. Ex 2a: $m = 105$
4. Ex 2b: No: they add to 180° (unless both are 90°)
5. Ex 3a: $p = 65$, $q = 115$
6. Ex 3b: corresponding (equal), alternate (equal), co-interior (add to 180°)

Easy · Set A

1. bottom ul
2. top lr
3. bottom ur
4. top ll
5. top lr
6. bottom ul

Easy · Set A2 (quick drill)

1. 60°
2. 105°
3. 85°
4. 120°
5. 35°
6. 150°
7. 70°
8. 92°

Easy · Set B

1. $a = 60$
2. $x = 110$
3. $b = 145$

Medium · Set C

1. $q = 62$
2. $k = 80$
3. $z = 123$ (co-interior 57° , then straight line)
4. $m = 64$ (corresponding)
5. $x = 130$
6. $y = 110$ (corresponding)

Challenging · Set D

1. $x = 60$
2. 90° each; it is perpendicular to the parallel lines
3. drawing
4. $x = 60$

Extension

1. 80° and 100°
2. $a = 50$

Exit ticket

1. Level 1: 70
2. Level 2: 180°
3. Level 3: $y = 20$

2.10 Finding unknown angles on parallel lines (booklet p. 57)

WE DO examples (teacher)

1. Ex 1a: $a = 65$ (corresponding then straight line; or co-interior with 115° pair)
2. Ex 1b: $m = 35$ (alternate)
3. Ex 2a: $x = 37$ (corr.), $y = 143$ (straight line)
4. Ex 2b: $k = 65$ (alternate)
5. Ex 3a: $p = 62$, $m = 62$, $z = 118$
6. Ex 3b: use two rules in steps, e.g. vertically opposite then alternate

Easy · Set A

1. $a = 65$
2. $b = 35$
3. $k = 65$
4. $d = 125$
5. $x = 50$
6. $q = 118$

Easy · Set A2 (quick drill)

1. corresponding, equal
2. alternate, equal
3. co-interior, add to 180°
4. corresponding, equal
5. alternate, equal
6. co-interior, add to 180°

Easy · Set B

1. $y = 132$ (alternate, then straight line)
2. $t = 110$
3. $x = 123$

Medium · Set C

1. $p = 62$, $m = 62$, $z = 118$
2. $t = 68$, $d = 112$, $a = 68$
3. $x = 130$, $y = 50$

Challenging · Set D

1. four of 57° and four of 123°
2. 135° and 45°
3. The lines are parallel so each angle has a matching corresponding angle
4. any correct question

Extension

1. $x = 16$
2. $x = 40$

Exit ticket

1. Level 1: 58
2. Level 2: alternate (vertically opposite then corresponding)
3. Level 3: four 58° and four 122°

2.11 Proving lines are parallel (booklet p. 63)**WE DO examples (teacher)**

1. Ex 1a: Yes: co-interior angles $135 + 45 = 180$
2. Ex 1b: No: $110 + 72 = 182$
3. Ex 2a: Yes: corresponding angles equal
4. Ex 2b: $x = 95$
5. Ex 3a: corresponding equal, alternate equal, co-interior supplementary
6. Ex 3b: we use the angle to prove the lines are parallel (the reverse of the rule)

Easy · Set A (quick drill)

1. yes
2. no
3. yes
4. no
5. yes
6. no
7. yes
8. yes
9. no
10. yes
11. no
12. yes

Easy · Set B

1. Yes (corresponding equal)
2. No (alternate $50 \neq 52$)
3. Yes (co-interior $115 + 65 = 180$)

Medium · Set C

1. 112
2. 47
3. 85
4. 55
5. 70
6. 37

Challenging · Set D

1. alternate angles equal, so parallel
2. $108 + 72 = 180$, co-interior supplementary, so parallel
3. No: corresponding angles are not equal
4. e.g. corresponding angles 60° and 70°

Extension

1. $x = 132$; $132 + 48 = 180$, so yes (co-interior)
2. Measurements are never exact: a proof uses the angle rules

Exit ticket

1. Level 1: Yes
2. Level 2: No: $105 + 85 = 190$, not 180
3. Level 3: $x = 37$